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Case Study 2: Architecture Design of the Proposed Work

Base Paper: CoWiC-MAC: A Coordinated WiFi-Centric MAC Mechanism in Heterogeneous Wireless Networks. Cheng, T., Li, S., and Pan, Y., Computer Networks, Vol. 287, Art. 112479, 2026

1. Overview

CoWiC-MAC is a coordination framework for WiFi and ZigBee devices that share the 2.4 GHz ISM band. Instead of letting the two technologies contend for the channel on their own, the WiFi Access Point acts as an orchestrator and decides when WiFi transmits and when ZigBee gets interference-free time. Coordination between the two sides happens over a Cross-Technology Communication (CTC) link that carries feedback in both directions. The architecture below shows the components and the flow between them.

2. Architecture Diagram

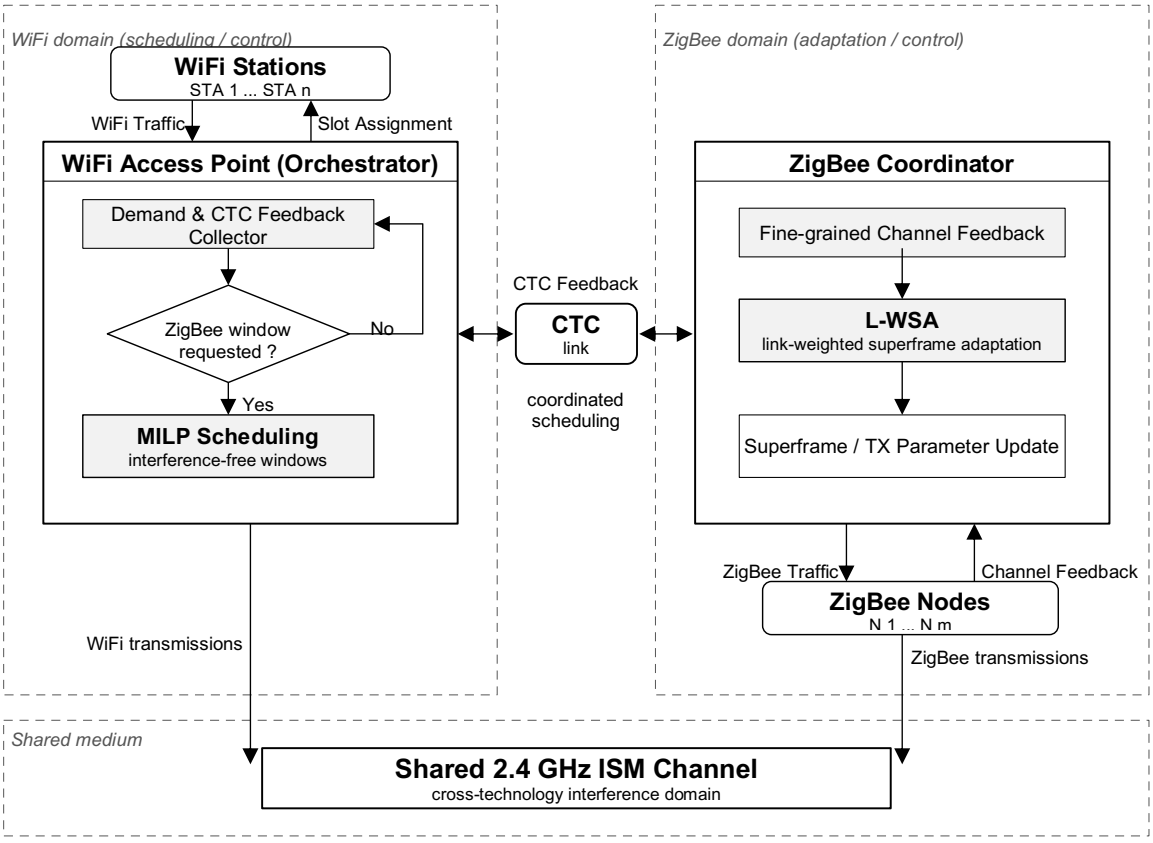


Fig. 1. Architecture of CoWiC-MAC

3. Component Description

WiFi Access Point (Orchestrator). The central component of the design. It has more computational capability than the low-power devices, so the scheduling work is placed here. It decides the order of WiFi transmissions and coordinates with the ZigBee side.

MILP-based Scheduling Model. Runs inside the AP. It uses Mixed-Integer Linear Programming to reorder WiFi traffic so that interference-free transmission opportunities are created for ZigBee. The model also keeps the scheduling overhead bounded, so reordering does not hurt WiFi.

CTC Link (Cross-Technology Communication). The communication path between the WiFi AP and the ZigBee coordinator. WiFi and ZigBee cannot normally talk to each other, so CTC provides the bidirectional feedback that makes coordinated scheduling possible.

ZigBee Coordinator. Manages the ZigBee side of the network. It receives coordination information over the CTC link and applies it to the ZigBee nodes under it.

L-WSA Algorithm (Link-Weighted Superframe Adaptation). A lightweight algorithm running at the ZigBee coordinator. It adapts the transmission parameters using fine-grained channel feedback. It is kept lightweight because ZigBee devices are resource constrained.

WiFi Stations. The normal WiFi client devices. Their traffic is transmitted in the order decided by the MILP schedule at the AP.

ZigBee Nodes. The low-power devices. They transmit during the interference-free windows created by the AP, using the parameters set by L-WSA.

Shared 2.4 GHz ISM Channel. The common medium used by both technologies. This is where cross-technology interference between high-power WiFi and low-power ZigBee occurs, and it is the resource the architecture is coordinating.

4. Data and Control Flow

1. The ZigBee coordinator and the WiFi AP exchange feedback over the CTC link. This makes the coordination demand driven instead of blind contention.

2. The MILP-based scheduling model at the AP reorders WiFi traffic and creates interference-free transmission opportunities for ZigBee, while bounding the scheduling overhead.

3. The AP sends the coordination information to the ZigBee coordinator over the CTC link.

4. The L-WSA algorithm at the ZigBee coordinator adapts the transmission parameters based on fine-grained channel feedback.

5. WiFi stations and ZigBee nodes then transmit on the shared 2.4 GHz ISM channel according to this coordinated schedule, which avoids the collisions caused by uncoordinated access.